# MARMA-CRYPTO & DSTU 4145-2002 GEOMETRIC SIGNATURE PROTOCOL

DSTU 4145-2002 Elliptic Curve GF(2^257) · GOST 34.311-95 · DSTU 7564 · 9-Triangle Marma Audit

Timestamp (UTC): 2026-10-17 05:05:20 UTC

VERIFIED: TRUE · DSTU 4145-2002 ELLIPTIC CURVE & 9-TRIANGLE SRI YANTRA GEOMETRY ARE VALID

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|---|---|
| Target File: | Marma_Crypto_Whitepaper_EN.pdf |
| Crypto Standard: | ···· 4145-2002 (··· ·····) / ···· ···· 34.311-95 / ···· 7564:2014 |
| Signer Subject (CN / RNOKPP): | ····· ····· ····· (RNOKPP: 3255419579) |
| Certificate Authority (QTSP): | ····· monobank \| Universal Bank |
| Key Medium & Fingerprint: | ·····_·_·_mono···_2025-09-15T20_12_28.pfx [b5913a2f269eb107db3ecf30...] |
| GOST 34.311-95 Digest: | 04611f0a87aef4fe0c0f2aee177b1544dc905cdd4ef5cbd063a964632569413b |
| DSTU 7564-256 (Kupyna): | 36b128ab98a4395bdb0e8efd15789b3d391b9f3dde42393968b5dbdf8ee65e46 |
| DSTU 4145-2002 Vector (r): | 0000000039765a188b8c84ced333a3adbc6bef21dc7dc2764a8c66da6d41bca4 |
| DSTU 4145-2002 Vector (s): | 000000000cbef919368df19c2fadfee017cd4d147a962af18470f8502b1b53d3 |
| Max Coordinate Drift (·_coord): | 0.0000000000 (threshold < 1e-8) |
| Bilateral Symmetry Error (·_sym): | 0.000000000000 (invariant lx + rx = 300.0) |

## 9-Triangle Micro-Perturbation Coordinate Audit Table (Bound to DSTU 4145-2002 Signature Vector r, s):

| Triangle | left_x (lx) | base_y (by) | apex_y (ay) | right_x (rx) | lx + rx (Sym) |
|---|---|---|---|---|---|
| D1 | 53.656704766 | 123.205085343 | 250.000000000 | 246.343295234 | 300.000000000 |
| U1 | 52.984019945 | 174.246610068 | 50.000000000 | 247.015980055 | 300.000000000 |
| U3 | 98.718230249 | 220.038288029 | 123.205080760 | 201.281769751 | 300.000000000 |
| U2 | 78.264677196 | 197.923155348 | 78.104991780 | 221.735322804 | 300.000000000 |
| D3 | 90.485691897 | 78.104991584 | 160.660149760 | 209.514308103 | 300.000000000 |
| D2 | 80.983848988 | 103.121991749 | 220.038289470 | 219.016151012 | 300.000000000 |
| U4 | 114.948848242 | 160.660148851 | 103.121991450 | 185.051151758 | 300.000000000 |
| D4 | 116.351418484 | 134.307572487 | 197.923156740 | 183.648581516 | 300.000000000 |
| D5 | 124.611906700 | 144.797771965 | 174.246605610 | 175.388093300 | 300.000000000 |